

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287537

Kind Code

A1

Publication Date

September 11, 2025

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THERMAL DIFFUSION DEVICE AND ELECTRONIC APPARATUS

Abstract

A vapor chamber **1**, which is an embodiment of a thermal diffusion device, includes: a housing **10** having a first inner surface **11a** and a second inner surface **12a** facing each other in a thickness direction Z, the housing **10** having an internal space; a working medium **20** enclosed in the internal space of the housing **10**; and a wick **30** disposed in the internal space of the housing **10**. The wick **30** has first through holes **61** having peripheries at which projections **65** toward the first inner surface **11a** of the housing **10** in the thickness direction Z are provided. A space **70** is provided between tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**. A maximum length of the space **70** in the thickness direction Z is smaller than an inner diameter of each of the first through holes **61**.

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Family ID: 91379352

Appl. No.: 19/218673

Filed: May 27, 2025

Foreign Application Priority Data

JP 2022-195644

Dec. 07, 2022

JP 2023-156426

Sep. 21, 2023

Related U.S. Application Data

parent WO continuation PCT/JP2023/042516 20231128 PENDING child US 19218673

Publication Classification

Int. Cl.: H05K7/20 (20060101)
U.S. Cl.:
CPC H05K7/20336 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International application No. PCT/JP2023/042516, filed Nov. 28, 2023, which claims priority to Japanese Patent Application No. 2022-195644, filed Dec. 7, 2022, and Japanese Patent Application No. 2023-156426, filed Sep. 21, 2023, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a thermal diffusion device and an electronic apparatus.

BACKGROUND ART

[0003] In recent years, elements have been highly integrated and have been improved in performance, thus increasing the amount of heat generated. In addition, size reductions of products increase the heat generation densities thereof. Thus, measures to dissipate heat are important. This situation is particularly conspicuous in the field of mobile terminals such as smart phones and tablets. For example, a graphite sheet is often used as a member for countermeasures against heat. However, the amount of heat transported thereby is not sufficient. Thus, use of various members for countermeasures against heat has been examined. In particular, use of vapor chambers, which are planar heat pipes and are thermal diffusion devices capable of diffusing heat significantly effectively, has been examined.

[0004] Such a vapor chamber has a structure in which a working medium (also referred to as a working liquid) and a wick configured to transport the working medium by capillary force are enclosed in a housing. The working medium absorbs heat from a heat generation element such as an electronic component in an evaporation portion configured to absorb heat from the heat generation element, evaporates in the vapor chamber, moves in the vapor chamber, and is cooled to return to a liquid phase. The working medium that has returned to the liquid phase is moved to the evaporation portion closer to the heat generation element again by the capillary force of the wick and cools the heat generation element. By repeating this operation, the vapor chamber is capable of operating autonomously without external power and of diffusing heat two-dimensionally at high speed by using evaporation latent heat and condensation latent heat of the working medium.

[0005] Patent Document 1 discloses a vapor chamber including a housing, a working liquid, a microchannel, and a sheet wick. The housing includes an upper housing sheet and a lower housing sheet joined at respective outer edge portions thereof and facing each other and has an internal space. The working liquid is enclosed in the internal space. The microchannel is disposed in the part of the internal space in the lower housing sheet and forms a passage for the working liquid. The wick is disposed in the internal space of the housing so as to be in contact with the microchannel. The area where the wick and the microchannel are in contact with each other is 5 percent to 40 percent of the area of the internal space in plan view. [0006] Patent Document 1: International Publication No. 2021/229961

SUMMARY OF THE DISCLOSURE

[0007] FIG. 1 of Patent Document 1 illustrates, as an embodiment of the vapor chamber, the structure in which the wick is interposed between projections of the microchannel formed in the lower housing sheet and pillars formed on the upper housing sheet. In addition, Patent Document 1

explains that the wick has a plurality of microscopic holes formed by, for example, etching.

[0008] The capillary force of the wick is determined by the diameter size of the holes provided in the wick. However, there is a limit to reducing the diameter of the holes due to the constraints of processing methods such as etching.

[0009] The above problem is not limited to the vapor chamber and is a problem common to thermal diffusion devices capable of diffusing heat with a configuration similar to that of the vapor chamber.

[0010] The present disclosure is made to solve the above problem, and an object of the present disclosure is to provide a thermal diffusion device having a high capillary force. In addition, another object of the present disclosure is to provide an electronic apparatus including the thermal diffusion device.

[0011] A thermal diffusion device according to the present disclosure includes: a housing having a first inner surface and a second inner surface facing each other in a thickness direction, the housing having an internal space; a working medium enclosed in the internal space of the housing; and a wick disposed in the internal space of the housing. The wick has a first through hole having a periphery at which a projection toward the first inner surface of the housing in the thickness direction is provided. A space is provided between a tip end surface of the projection and the first inner surface of the housing. A maximum length of the space in the thickness direction is smaller than an inner diameter of the first through hole.

[0012] An electronic apparatus according to the present disclosure includes the thermal diffusion device according to the present disclosure.

[0013] According to the present disclosure, it is possible to provide a thermal diffusion device having a high capillary force. In addition, according to the present disclosure, it is possible to provide an electronic apparatus including the thermal diffusion device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a schematic perspective view of an example of a thermal diffusion device according to a first embodiment of the present disclosure.

[0015] FIG. 2 is a schematic sectional view of an example of the thermal diffusion device according to the first embodiment of the present disclosure.

[0016] FIG. 3 is a schematic sectional view of examples of a housing and a wick that form the thermal diffusion device according to the first embodiment of the present disclosure.

[0017] FIG. 4 is a schematic plan view of an example of the wick illustrated in FIG. 3.

[0018] FIG. 5 is a schematic perspective view of an example of a projection provided at the periphery of a first through hole.

[0019] FIG. 6 is a schematic sectional view of a first modification example of a part represented by a in FIG. 3.

[0020] FIG. 7 is a schematic sectional view of a second modification example of the part represented by a in FIG. 3.

[0021] FIG. 8 is a schematic sectional view of a third modification example of the part represented by a in FIG. 3.

[0022] FIG. 9 is a schematic sectional view of a fourth modification example of the part represented by a in FIG. 3.

[0023] FIGS. 10A and 10B are schematic perspective views illustrating an example of a method for forming folded protrusions at a tip end surface of the projection.

[0024] FIG. 11 is a schematic sectional view of a fifth modification example of the part represented by a in FIG. 3.

[0025] FIG. 12 is a schematic sectional view of a sixth modification example of the part represented by a in FIG. 3.

[0026] FIG. 13 is a schematic sectional view of a seventh modification example of the part represented by a in FIG. 3.

[0027] FIG. 14 is a schematic sectional view of examples of a housing and a wick that form a thermal diffusion device according to a second embodiment of the present disclosure.

[0028] FIG. 15 is a schematic sectional view of a modification example of the shape of the projection toward a first inner surface of the housing.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0029] A thermal diffusion device according to the present disclosure will be described below.

[0030] However, the present disclosure is not limited to the following embodiments, and modifications can be applied thereto as appropriate without changing the gist of the present disclosure. Combinations of two or more different preferable configurations of the present disclosure described below are also included in the present disclosure.

[0031] In the thermal diffusion device according to the present disclosure, a wick has first through holes having peripheries at which projections toward a first inner surface of a housing in the thickness direction are provided. Thus, it is possible to form a liquid passage for a working medium between the projections adjacent to each other.

[0032] In addition, in the thermal diffusion device according to the present disclosure, a space is provided between tip end surfaces of the projections provided at the peripheries of the first through holes and the first inner surface of the housing such that the first through holes are not closed by the first inner surface of the housing. This space causes a capillary force. In addition, the maximum length of the space in the thickness direction (hereinafter also referred to as the height of the space) is smaller than the inner diameter of each of the first through holes, thus causing a higher capillary force.

[0033] As described in Patent Document 1, there is a limit to reducing the diameter of the holes of the wick due to processing methods. However, the structure in which the space is provided between the tip end surfaces of the projections and the first inner surface of the housing facilitates the control of the height of the space. Thus, it is possible to achieve an effect similar to the effect of reducing the diameter of the holes.

[0034] It is needless to say that the embodiments described below are examples, and configurations of different embodiments can be partially replaced or combined. From a second embodiment, points in common with a first embodiment are not described, and only different points are described. In particular, similar operational effects resulting from similar configurations are not mentioned in each embodiment.

[0035] In the following description, when the embodiments are not particularly distinguished from each other, it is simply referred to as “thermal diffusion device according to the present disclosure”.

[0036] In the following description, a vapor chamber is taken as an embodiment of the thermal diffusion device according to the present disclosure. The thermal diffusion device according to the present disclosure is also applicable to thermal diffusion devices such as a heat pipe.

[0037] Figures described below are schematic, and, for example, the size or the aspect ratio thereof may differ from that of an actual product. In the figures, the same reference signs are used for the same or corresponding portions. In addition, in the figures, the same elements have the same reference signs, and redundant descriptions thereof are omitted.

[0038] In the description, the terms representing the relationships between elements (such as “vertical”, “parallel”, and “orthogonal”) and the terms representing the shapes of the elements not only have strictly literal meanings but also have substantially equivalent meanings, for example, with a difference of about several percent.

First Embodiment

[0039] In the thermal diffusion device according to the first embodiment of the present disclosure,

in addition to the first through holes having the peripheries at which the projections toward the first inner surface of the housing in the thickness direction are provided, the wick has second through holes having peripheries at which no projections toward the first inner surface or a second inner surface of the housing in the thickness direction are provided.

[0040] FIG. 1 is a schematic perspective view of an example of the thermal diffusion device according to the first embodiment of the present disclosure. FIG. 2 is a schematic sectional view of an example of the thermal diffusion device according to the first embodiment of the present disclosure. FIG. 2 is an example of a sectional view of the thermal diffusion device taken along line II-II in FIG. 1.

[0041] A vapor chamber (thermal diffusion device) **1** illustrated in FIGS. 1 and 2 includes a housing **10**, which is hollow and is hermetically sealed. The housing **10** has a first inner surface **11a** and a second inner surface **12a**, which face each other in a thickness direction Z. The housing **10** has an internal space. The vapor chamber **1** further includes a working medium **20** enclosed in the internal space of the housing **10**, and a wick **30** disposed in the internal space of the housing **10**. The vapor chamber **1** may further include pillars **40** disposed in the internal space of the housing **10**.

[0042] An evaporation portion configured to evaporate the enclosed working medium **20** is set in the housing **10**. As illustrated in FIG. 1, a heat source HS, which is a heat generation element, is disposed on an outer surface of the housing **10**. Examples of the heat source HS include electronic components of an electronic apparatus, such as a central processing unit (CPU). A part in the internal space of the housing **10** that is located in the vicinity of the heat source HS and that is configured to be heated by the heat source HS corresponds to the evaporation portion.

[0043] The vapor chamber **1** preferably has a planar shape as a whole. That is, the housing **10** preferably has a planar shape as a whole. Here, “planar shape” includes a plate-like shape and a sheet-like shape and means a shape in which the measurement in a width direction X (hereinafter referred to as the width) and the measurement in a length direction Y (hereinafter referred to as the length) are considerably larger than the measurement in the thickness direction Z (hereinafter referred to as the thickness or the height), for example, a shape in which the width and the length are 10 times or more, preferably 100 times or more, of the thickness.

[0044] The size of the vapor chamber **1**, that is, the size of the housing **10** is not particularly limited. The width and the length of the vapor chamber **1** can be appropriately set according to its use. Each of the width and the length of the vapor chamber **1** is, for example, 5 mm to 500 mm, 20 mm to 300 mm, or 50 mm to 200 mm. The width and the length of the vapor chamber **1** may be equal to or different from each other.

[0045] The housing **10** is preferably formed by a first sheet **11** and a second sheet **12**, which are joined at respective outer edge portions thereof and face each other.

[0046] When the housing **10** is formed by the first sheet **11** and the second sheet **12**, the material forming the first sheet **11** and the second sheet **12** is not particularly limited as long as having characteristics, such as a thermal conductivity, a strength, elasticity, and flexibility, suitable for such a thermal diffusion device such as a vapor chamber. The material forming the first sheet **11** and the second sheet **12** is preferably a metal. Examples of such a metal include copper, nickel, aluminum, magnesium, titanium, iron, and alloys mainly composed of these metals. Particularly preferably, the material forming the first sheet **11** and the second sheet **12** is copper. The material forming the first sheet **11** may be the same as or different from the material forming the second sheet **12**. Preferably, the material forming the first sheet **11** is the same as the material forming the second sheet **12**.

[0047] When the housing **10** is formed by the first sheet **11** and the second sheet **12**, the first sheet **11** and the second sheet **12** are joined at the respective outer edge portions thereof. The joining method is not particularly limited. Usable examples of such a joining method include laser welding, resistance welding, diffusion bonding, brazing, TIG welding (tungsten-inert gas welding),

ultrasonic welding, and resin sealing. Preferably, it is possible to use laser welding, resistance welding, or brazing.

[0048] The thickness of the first sheet **11** and the thickness of the second sheet **12** are not particularly limited. Each of the thickness of the first sheet **11** and the thickness of the second sheet **12** is preferably 10 μm to 200 μm , more preferably 30 μm to 100 μm , even more preferably 40 μm to 60 μm . The thickness of the first sheet **11** and the thickness of the second sheet **12** may be equal to or different from each other. In addition, the thickness of each of the first sheet **11** and the second sheet **12** may be entirely uniform or partially thin.

[0049] The shape of the first sheet **11** and the shape of the second sheet **12** are not particularly limited. For example, each of the first sheet **11** and the second sheet **12** may be shaped such that the outer edge portion is thicker than the part thereof other than the outer edge portion.

[0050] The thickness of the entire vapor chamber **1** is not particularly limited and is preferably 50 μm to 500 μm .

[0051] The shape of the housing **10** in plan view when viewed in the thickness direction Z is not particularly limited. Examples of the shape of the housing **10** in plan view include a polygon such as a triangle or a rectangle, a circle, an ellipse, and shapes formed by combining these shapes. In addition, the shape of the housing **10** in plan view may be, for example, an L shape, a C shape (U shape), or a stepped shape. In addition, the housing **10** may have a through hole. The shape of the housing **10** in plan view may be a shape formed according to the use of a thermal diffusion device such as a vapor chamber, the shape of a part into which the thermal diffusion device is inserted, or other components located in the vicinity of the housing **10**.

[0052] The working medium **20** is not particularly limited as long as being capable of changing between a gas phase and a liquid phase in the environment in the housing **10**. Usable examples of the working medium **20** include water, alcohol, and alternative CFCs. For example, the working medium **20** is an aqueous compound, preferably water.

[0053] The wick **30** has a capillary structure capable of moving the working medium **20** by capillary force. The wick **30** preferably has a planar shape.

[0054] The material forming the wick **30** is not particularly limited and is preferably a metal. Examples of such a metal include copper, nickel, aluminum, magnesium, titanium, iron, and alloys mainly composed of these metals. Particularly preferably, the material forming the wick **30** is copper. The material forming the wick **30** may be the same as or different from the material forming the housing **10**.

[0055] The size and the shape of the wick **30** are not particularly limited. For example, the wick **30** is preferably disposed continuously in the internal space of the housing **10**. When viewed in the thickness direction Z, the wick **30** may be disposed in the entire internal space of the housing **10**. When viewed in the thickness direction Z, the wick **30** may be disposed in part of the internal space of the housing **10**.

[0056] The thickness of the wick **30** is not particularly limited and is, for example, 5 μm to 50 μm .

[0057] As illustrated in FIG. 2, the pillars **40** in contact with the second inner surface **12a** may be disposed in the internal space of the housing **10**. The disposition of the pillars **40** in the internal space of the housing **10** enables the housing **10** and the wick **30** to be supported.

[0058] The material forming the pillars **40** is not particularly limited. Examples of the material forming the pillars **40** include a resin, a metal, ceramic, and mixtures and laminates of these substances. In addition, as illustrated in FIG. 2, the pillars **40** may be integrally formed with the housing **10** and may be formed by, for example, subjecting the second inner surface **12a** of the housing **10** to etching.

[0059] The shape of the pillars **40** is not particularly limited as long as the pillars **40** are shaped so as to be able to support the housing **10** and the wick **30**. Examples of the shape of a section of the pillar **40** perpendicular to the height direction include a polygon such as a rectangle, a circle, and an ellipse.

[0060] As illustrated in FIG. 2, the pillar **40** may have a tapered shape in which the width of the pillar **40** narrows from the second inner surface **12a** of the housing **10** toward the wick **30**. Thus, it is possible to widen the part of a passage between the pillars **40** on the wick **30** side.

[0061] In the vapor chamber, each height of the pillars **40** may be equal or different. The height of the pillars **40** is, for example, 50 μm to 1000 μm .

[0062] The disposition of the pillars **40** is not particularly limited. The pillars **40** in a predetermined region are preferably evenly disposed, or more preferably all the pillars **40** are evenly disposed such that, for example, the center-to-center distance (pitch) of the pillars **40** adjacent to each other is uniform. The pillars **40** are evenly disposed, thus enabling the entire thermal diffusion device such as a vapor chamber to have a uniform strength. The center-to-center distance of the pillars **40** is, for example, 100 μm to 5000 μm .

[0063] In the section illustrated in FIG. 2, the width of the pillar **40** is not particularly limited as long as the pillar **40** has a strength sufficient to inhibit deformation of the housing **10**. The equivalent circle diameter of a section perpendicular to the height direction of the end portion of the pillar **40** closer to the wick **30** is, for example, 100 μm to 2000 μm , preferably 300 μm to 1000 μm . An increase in the equivalent circle diameter of the pillar **40** enables deformation of the housing **10** to be further inhibited. On the other hand, a reduction in the equivalent circle diameter of the pillar **40** enables formation of a wider space for moving vapor of the working medium **20**.

[0064] In the vapor chamber **1**, the wick **30** has through holes **60** passing through the wick **30** in the thickness direction Z.

[0065] Capillary action enables the working medium **20** to move in the through holes **60**. The shape of the through holes **60** is not particularly limited. Preferably, a section of the through hole **60** perpendicular to the thickness direction Z is a circle or an ellipse.

[0066] The disposition of the through holes **60** is not particularly limited. The through holes **60** in a predetermined region are preferably evenly disposed, or more preferably all the through holes **60** are evenly disposed such that, for example, the center-to-center distance (pitch) of the through holes **60** adjacent to each other is uniform.

[0067] The through holes **60** can be formed by, for example, punching metal foil forming the wick **30** in press working.

[0068] FIG. 3 is a schematic sectional view of examples of a housing and a wick that form the thermal diffusion device according to the first embodiment of the present disclosure. FIG. 4 is a schematic plan view of an example of the wick illustrated in FIG. 3. FIG. 3 illustrates a section of the wick taken along line A-A in FIG. 4.

[0069] In the example illustrated in FIGS. 3 and 4, the wick **30** has first through holes **61** and second through holes **62**, which are the through holes **60**.

[0070] As illustrated in FIG. 3, a projection **65** toward the first inner surface **11a** of the housing **10** in the thickness direction Z is provided at the periphery of each first through hole **61**.

[0071] On the other hand, no projection **65** toward the first inner surface **11a** or the second inner surface **12a** (see FIG. 2) of the housing **10** in the thickness direction Z is provided at the periphery of each second through hole **62**.

[0072] The working medium **20** in a liquid phase is held between the projections **65** provided at the peripheries of the first through holes **61**. Thus, it is possible to improve the heat transport performance of a thermal diffusion device such as a vapor chamber.

[0073] In addition, a space **70** is provided between tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**. The maximum length of the space **70** in the thickness direction Z (the height of the space **70**) is smaller than the inner diameter of each of the first through holes **61**. The space **70** causes a high capillary force.

[0074] The maximum length of the space **70** in the thickness direction Z may be, for example, 30 μm or less. On the other hand, the maximum length of the space **70** in the thickness direction Z may be, for example, 1 μm or more.

[0075] The inner diameter of each of the first through holes **61** may be, for example, 40 μm or less. On the other hand, the inner diameter of each of the first through holes **61** may be, for example, 5 μm or more.

[0076] The inner diameter of the first through hole **61** means the equivalent circle diameter of the first through hole **61** located in the tip end surface **65a** of the projection **65**. The inner diameter of the first through hole **61** is preferably smaller than the equivalent circle diameter of a section perpendicular to the height direction of the end portion of the pillar **40** closer to the wick **30**.

[0077] The center-to-center distance of the first through holes **61** is, for example, 60 μm to 800 μm . The center-to-center distance of the first through holes **61** is preferably smaller than the center-to-center distance of the pillars **40**.

[0078] The inner diameter of each of the second through holes **62** is, for example, 100 μm or less. The inner diameter of the second through hole **62** may be equal to the inner diameter of the first through hole **61**, may be larger than the inner diameter of the first through hole **61**, or may be smaller than the inner diameter of the first through hole **61**. When the second through holes **62** have different inner diameters when viewed in the thickness direction Z, the inner diameter of the smallest one of the second through holes **62** is defined as the inner diameter of the second through hole **62**.

[0079] The center-to-center distance of the second through holes **62** is, for example, 3 μm to 150 μm . The center-to-center distance of the second through holes **62** may be equal to the center-to-center distance of the first through holes **61**, may be larger than the center-to-center distance of the first through holes **61**, or may be smaller than the center-to-center distance of the first through holes **61**.

[0080] The disposition of the projections **65** is not particularly limited. The projections **65** in a predetermined region are preferably evenly disposed, or more preferably all the projections **65** are evenly disposed such that, for example, the center-to-center distance (pitch) of the projections **65** adjacent to each other is uniform.

[0081] The height of the projection **65** is, for example, 10 μm to 100 μm . The height of the projection **65** is preferably smaller than the height of the pillar **40**.

[0082] The size of the projection **65** is not particularly limited. For example, the height of the projection **65** may be larger than the inner diameter of the first through hole **61**, may be smaller than the inner diameter of the first through hole **61**, or may be equal to the inner diameter of the first through hole **61**.

[0083] The projections **65** can be formed by, for example, punching metal foil forming the wick **30** in press working. In this case, the projections **65** may be formed simultaneously with the through holes **60** such as the first through holes **61** or may be formed separately from the through holes **60** such as the first through holes **61**. In the punching process in press working, for example, the shape of the projections **65** can be adjusted by appropriately adjusting the punching depth or the like. The punching depth means, for example, the depth of a punch pushed in a punching direction when the punching process is performed by using the punch.

[0084] The thickness of metal foil yet to be subjected to working such as press working is preferably uniform. However, parts of the metal foil subjected to press working may be thin. Thus, in the example illustrated in FIG. 3, preferably, the thickness of the projections **65** is equal to the thickness of the part of the wick **30** other than the projections **65** or is smaller than the thickness of the part of the wick **30** other than the projections **65**.

[0085] FIG. 5 is a schematic perspective view of an example of a projection provided at the periphery of a first through hole.

[0086] In the example illustrated in FIG. 5, the tip end surface **65a** of the projection **65** has a flat structure.

[0087] The projection **65** may be provided only at part of the periphery of the first through hole **61**. However, as illustrated in FIG. 5, the projection **65** is preferably provided at the entire periphery of

the first through hole **61**.

[0088] Preferably, the projections **65** are not joined to the first inner surface **11a** of the housing **10**. As long as the space **70** is provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**, part of the tip end surface **65a** of the projection **65** may be in contact with the first inner surface **11a** of the housing **10**.

[0089] When a plurality of the projections **65** are provided to the wick **30**, the shape of the space **70** may be a uniform shape, a shape formed by combining two or more shapes, or an entirely randomly varying shape. In addition, the projections **65** may be included such that no space **70** is provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**.

[0090] The tip end surface **65a** of the projection **65** may have an uneven structure. In this case, the tip end surface **65a** of the projection **65** having the uneven structure may enable the space **70** to be provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**.

[0091] FIG. **6** is a schematic sectional view of a first modification example of a part represented by a in FIG. **3**. In FIG. **3**, a represents the part between the tip end surface **65a** of the projection **65** and the first inner surface **11a** of the housing **10**. The same applies to FIG. **14** to be described later.

[0092] In the example illustrated in FIG. **6**, the tip end surface **65a** of the projection **65** has a curvilinear wave sectional shape.

[0093] FIG. **7** is a schematic sectional view of a second modification example of the part represented by a in FIG. **3**.

[0094] In the example illustrated in FIG. **7**, the tip end surface **65a** of the projection **65** has a triangular wave sectional shape.

[0095] FIG. **8** is a schematic sectional view of a third modification example of the part represented by a in FIG. **3**.

[0096] In the example illustrated in FIG. **8**, the tip end surface **65a** of the projection **65** has a rectangular wave sectional shape.

[0097] FIG. **9** is a schematic sectional view of a fourth modification example of the part represented by a in FIG. **3**.

[0098] In the example illustrated in FIG. **9**, folded protrusions **65b** each protruding toward the inside of the first through hole **61** in a radial direction are provided to the tip end surface **65a** of the projection **65**.

[0099] FIGS. **10A** and **10B** are schematic perspective views illustrating an example of a method for forming folded protrusions at the tip end surface of the projection.

[0100] For example, when the projections **65** are formed by punching in working such as press working, as illustrated in FIG. **10A**, plate protrusions **65c** are formed on the tip end surface **65a** of the projection **65** as burrs formed during the working. Thereafter, the projections **65** are flattened by working such as roll pressing. Thus, as illustrated in FIG. **10B**, the plate protrusions **65c** are bent to form the folded protrusions **65b**.

[0101] Alternatively, the first inner surface **11a** of the housing **10** may have an uneven structure, thus enabling the space **70** to be provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**. In this case, the tip end surface **65a** of the projection **65** may have a flat structure or an uneven structure.

[0102] FIG. **11** is a schematic sectional view of a fifth modification example of the part represented by a in FIG. **3**.

[0103] In the example illustrated in FIG. **11**, the first inner surface **11a** of the housing **10** has a curvilinear wave sectional shape.

[0104] FIG. **12** is a schematic sectional view of a sixth modification example of the part represented by a in FIG. **3**.

[0105] In the example illustrated in FIG. **12**, the first inner surface **11a** of the housing **10** has a

triangular wave sectional shape.

[0106] FIG. **13** is a schematic sectional view of a seventh modification example of the part represented by a in FIG. **3**.

[0107] In the example illustrated in FIG. **13**, the first inner surface **11a** of the housing **10** has a rectangular wave sectional shape.

[0108] Any of the structures of the tip end surface **65a** of the projection **65** illustrated in FIGS. **6** to **9** may be combined with any of the structures of the first inner surface **11a** of the housing **10** illustrated in FIGS. **11** to **13**.

Second Embodiment

[0109] In the thermal diffusion device according to the second embodiment of the present disclosure, the wick only has the first through holes having peripheries at which the projections toward the first inner surface of the housing in the thickness direction are provided.

[0110] FIG. **14** is a schematic sectional view of examples of the housing and the wick that form the thermal diffusion device according to the second embodiment of the present disclosure.

[0111] In the example illustrated in FIG. **14**, the wick **30** only has the first through holes **61**, which are the through holes **60**. On the other hand, the wick **30** has no second through holes **62** (see FIG. **3**), which are the through holes **60**.

[0112] As illustrated in FIG. **14**, the projections **65** toward the first inner surface **11a** of the housing **10** in the thickness direction Z are provided at the peripheries of the first through holes **61**.

[0113] In addition, the space **70** is provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**. The maximum length of the space **70** in the thickness direction Z (the height of the space **70**) is smaller than the inner diameter of the first through hole **61**.

[0114] In addition, the structure of a part represented by a in FIG. **14** is similar to the structure described in the first embodiment.

OTHER EMBODIMENTS

[0115] The thermal diffusion device according to the present disclosure is not limited to the above embodiments. Various applications and modifications can be made to, for example, the configuration and the manufacturing conditions of the thermal diffusion device within the scope of the present disclosure.

[0116] For example, the shape of the projections **65** toward the first inner surface **11a** of the housing **10** in the thickness direction Z is not particularly limited.

[0117] FIG. **15** is a schematic sectional view of a modification example of the shape of the projection toward the first inner surface of the housing.

[0118] As illustrated in FIG. **15**, the distance across an outer wall of the projection **65** may be uniform toward the tip end surface **65a** of the projection **65**.

[0119] Alternatively, as illustrated in, for example, FIG. **3**, the distance across the outer wall of the projection **65** may be reduced toward the tip end surface **65a** of the projection **65**. In this case, the projection **65** may be shaped so as to project toward the first inner surface **11a** (lower side in FIG. **3**) or the second inner surface **12a** (upper side in FIG. **3**).

[0120] In addition, the distance across the outer wall of the projection **65** may be increased toward the tip end surface **65a** of the projection **65**. In this case, the projection **65** may be shaped so as to project toward the first inner surface **11a** or the second inner surface **12a**.

[0121] When the wick **30** has a plurality of the first through holes **61**, the space **70** having a height (the maximum length in the thickness direction Z) larger than the inner diameter of the first through hole **61** may be provided between the tip end surfaces **65a** of the projections **65** and the first inner surface **11a** of the housing **10**.

[0122] When the wick **30** has a plurality of the through holes **60**, the first through holes **61** are included in the through holes **60**, and the second through holes **62** may be included in the through holes **60** or do not have to be included in the through holes **60**.

[0123] In the thermal diffusion device according to the present disclosure, the housing may include one or a plurality of evaporation portions. That is, one or a plurality of heat sources may be disposed on the outer wall surface of the housing.

[0124] In the thermal diffusion device according to the present disclosure, when the housing is formed by the first sheet and the second sheet, the first sheet and the second sheet may overlap such that end portions thereof coincide or do not coincide with each other.

[0125] In the thermal diffusion device according to the present disclosure, when the housing is formed by the first sheet and the second sheet, the material forming the first sheet and the material forming the second sheet may be different from each other. For example, use of a material having a high strength for the first sheet enables the stress applied to the housing to be dispersed. In addition, use of different materials for the respective sheets enables one of the sheets to have one function and the other of the sheets to have another function. Such functions are not particularly limited. Examples of such functions include a thermal conduction function and an electromagnetic shielding function.

[0126] The thermal diffusion device according to the present disclosure can be mounted in an electronic apparatus to dissipate heat. Thus, the present disclosure also includes an electronic apparatus including the thermal diffusion device according to the present disclosure. Examples of the electronic apparatus according to the present disclosure include a smart phone, a tablet terminal, a notebook computer, a game device, and a wearable device. As described above, the thermal diffusion device according to the present disclosure is capable of operating autonomously without external power and of diffusing heat two-dimensionally at high speed by using evaporation latent heat and condensation latent heat of a working medium. Thus, the electronic apparatus including the thermal diffusion device according to the present disclosure is capable of effectively dissipating heat in a limited space in the electronic apparatus.

[0127] The description discloses the following contents.

[0128] <1> A thermal diffusion device comprising: a housing having a first inner surface and a second inner surface facing each other in a thickness direction, the housing having an internal space; a working medium enclosed in the internal space of the housing; and a wick disposed in the internal space of the housing, wherein the wick has a first through hole having a periphery at which a projection toward the first inner surface of the housing in the thickness direction is provided, a space is provided between a tip end surface of the projection and the first inner surface of the housing, and a maximum length of the space in the thickness direction is smaller than an inner diameter of the first through hole.

[0129] <2> The thermal diffusion device according to <1>, wherein the tip end surface of the projection has an uneven structure to provide the space between the tip end surface of the projection and the first inner surface of the housing.

[0130] <3> The thermal diffusion device according to <1> or <2>, wherein the first inner surface of the housing has an uneven structure to provide the space between the tip end surface of the projection and the first inner surface of the housing.

[0131] <4> The thermal diffusion device according to any one of <1> to <3>, wherein the wick further has a second through hole having a periphery at which no projection toward the first inner surface or the second inner surface of the housing in the thickness direction is provided.

[0132] <5> The thermal diffusion device according to any one of <1> to <4>, wherein the maximum length of the space in the thickness direction is 30 μm or less.

[0133] <6> The thermal diffusion device according to any one of <1> to <5>, wherein the inner diameter of the first through hole is 40 μm or less.

[0134] <7> The thermal diffusion device according to any one of <1> to <6>, wherein the projection is not joined to the first inner surface of the housing.

[0135] <8> The thermal diffusion device according to any one of <1> to <7>, wherein a distance across an outer wall of the projection is reduced toward the tip end surface of the projection.

[0136] <9> The thermal diffusion device according to any one of <1> to <7>, wherein a distance across an outer wall of the projection is uniform toward the tip end surface of the projection.

[0137] <10> An electronic apparatus comprising the thermal diffusion device according to any one of <1> to <9>.

[0138] The thermal diffusion device according to the present disclosure can be used for various purposes in the field of, for example, portable information terminals. For example, the thermal diffusion device according to the present disclosure can be used to reduce the temperature of a heat source such as a CPU and to extend the operating time of an electronic apparatus and can be used for smart phones, tablet terminals, notebook computers, and the like.

REFERENCE SIGNS LIST

[0139] **1** vapor chamber (thermal diffusion device) [0140] **10** housing [0141] **11** first sheet [0142] **11a** first inner surface [0143] **12** second sheet [0144] **12a** second inner surface [0145] **20** working medium [0146] **30** wick [0147] **40** pillar [0148] **60** through hole [0149] **61** first through hole [0150] **62** second through hole [0151] **65** projection [0152] **65a** tip end surface [0153] **65b** folded protrusion [0154] **65c** plate protrusion [0155] **70** space [0156] HS heat source [0157] X width direction [0158] Y length direction [0159] Z thickness direction

Claims

1. A thermal diffusion device comprising: a housing having a first inner surface and a second inner surface facing each other in a thickness direction and defining an internal space; a working medium enclosed in the internal space of the housing; and a wick in the internal space of the housing, wherein the wick has a first through hole having a first periphery and a projection at the first periphery of the through hole, the projection extending toward the first inner surface of the housing in the thickness direction so as to define a space between a tip end surface of the projection and the first inner surface of the housing, and a maximum length of the space in the thickness direction is smaller than an inner diameter of the first through hole.
2. The thermal diffusion device according to claim 1, wherein the tip end surface of the projection has an uneven structure.
3. The thermal diffusion device according to claim 1, wherein the tip end surface of the projection has a flat structure.
4. The thermal diffusion device according to claim 1, wherein the first inner surface of the housing has an uneven structure.
5. The thermal diffusion device according to claim 1, wherein the wick further has a second through hole having a second periphery.
6. The thermal diffusion device according to claim 5, wherein the second periphery does not have a projection extending toward the first inner surface or the second inner surface of the housing in the thickness direction.
7. The thermal diffusion device according to claim 5, wherein an inner diameter of the second through hole is 100 μm or less.
8. The thermal diffusion device according to claim 1, wherein the maximum length of the space in the thickness direction is 30 μm or less.
9. The thermal diffusion device according to claim 1, wherein the maximum length of the space in the thickness direction is 1 μm to 30 μm .
10. The thermal diffusion device according to claim 1, wherein the inner diameter of the first through hole is 40 μm or less.
11. The thermal diffusion device according to claim 1, wherein the inner diameter of the first through hole is 5 μm to 40 μm .
12. The thermal diffusion device according to claim 1, wherein the projection is not joined to the first inner surface of the housing.

- 13.** The thermal diffusion device according to claim 1, wherein a distance across an outer wall of the projection reduces in a direction toward the tip end surface of the projection.
- 14.** The thermal diffusion device according to claim 1, wherein a distance across an outer wall of the projection is uniform in a direction toward the tip end surface of the projection.
- 15.** The thermal diffusion device according to claim 1, wherein the thermal diffusion device includes a plurality of the first through holes and a center-to-center distance of adjacent first through holes of the plurality of first through holes is 60 μm to 800 μm .
- 16.** The thermal diffusion device according to claim 15, further comprising a plurality of pillars extending in the thickness direction from the second inner surface, and wherein the center-to-center distance of the adjacent first through holes is smaller than a center-to-center distance of adjacent pillars of the plurality of pillars.
- 17.** The thermal diffusion device according to claim 15, wherein the thermal diffusion device includes a plurality of second through holes and a center-to-center distance of adjacent second through holes of the plurality of second through holes is 3 μm to 150 μm .
- 18.** The thermal diffusion device according to claim 1, wherein a height of the projection in the thickness direction is 10 μm to 100 μm .
- 19.** An electronic apparatus comprising the thermal diffusion device according to claim 1.
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